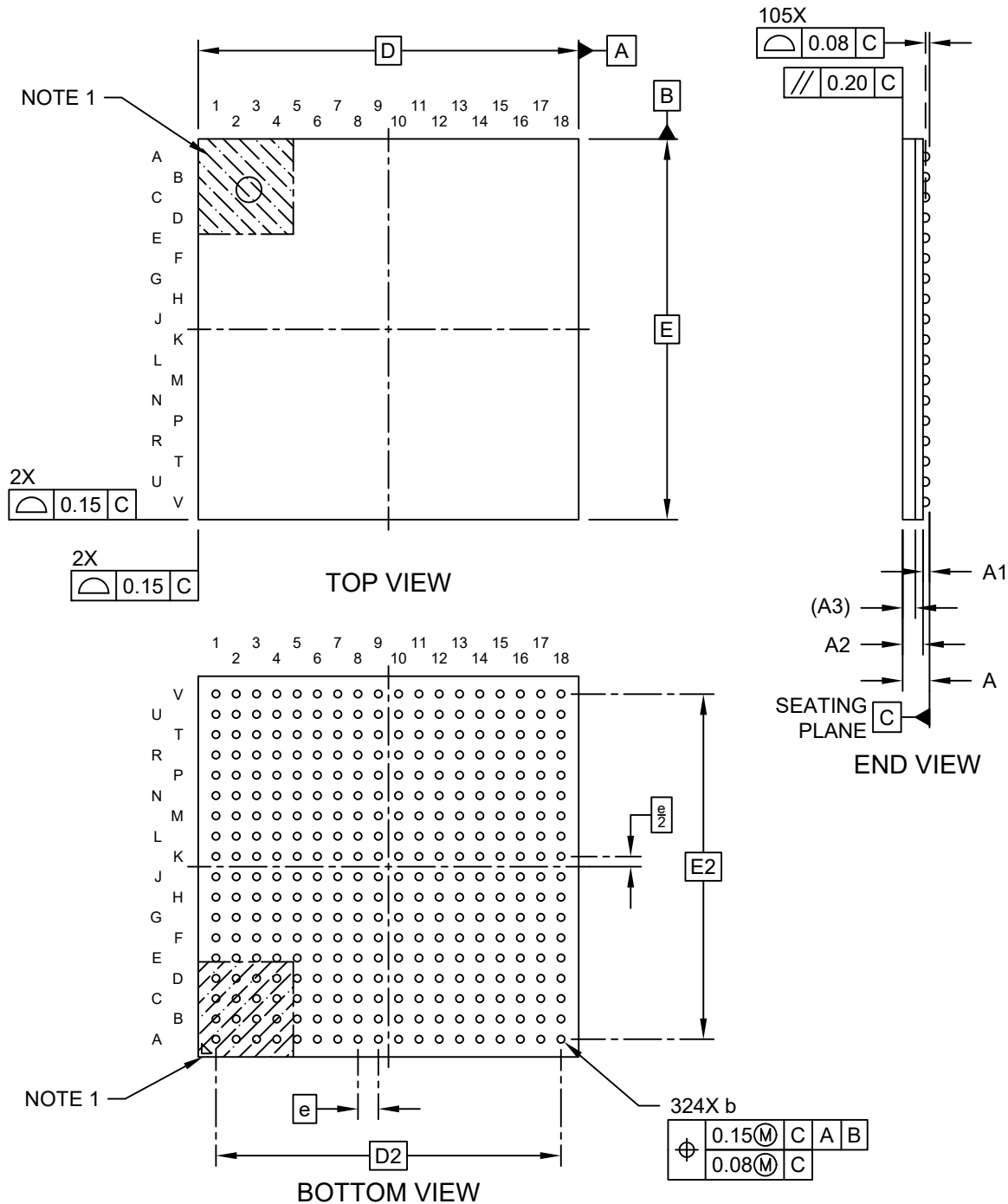


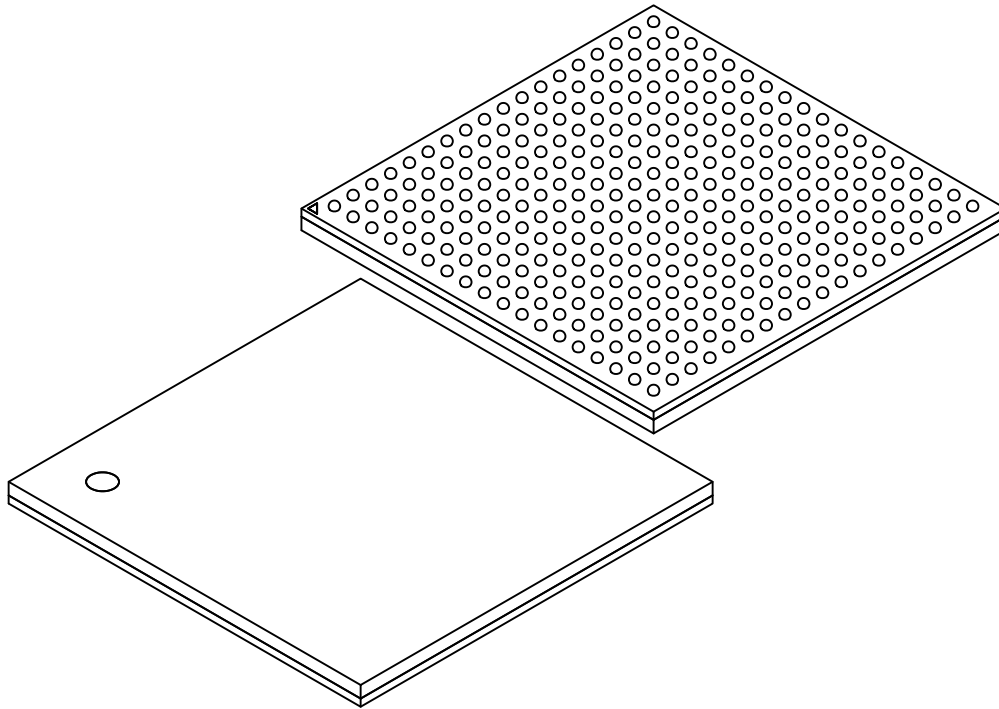
324-Ball Thin Fine-Pitch Ball Grid Array (BJB) - 15x15x1.2 mm Body [TFBGA] Atmel Legacy Package CRC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



324-Ball Thin Fine-Pitch Ball Grid Array (BJB) - 15x15x1.2 mm Body [TFBGA] Atmel Legacy Package CRC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



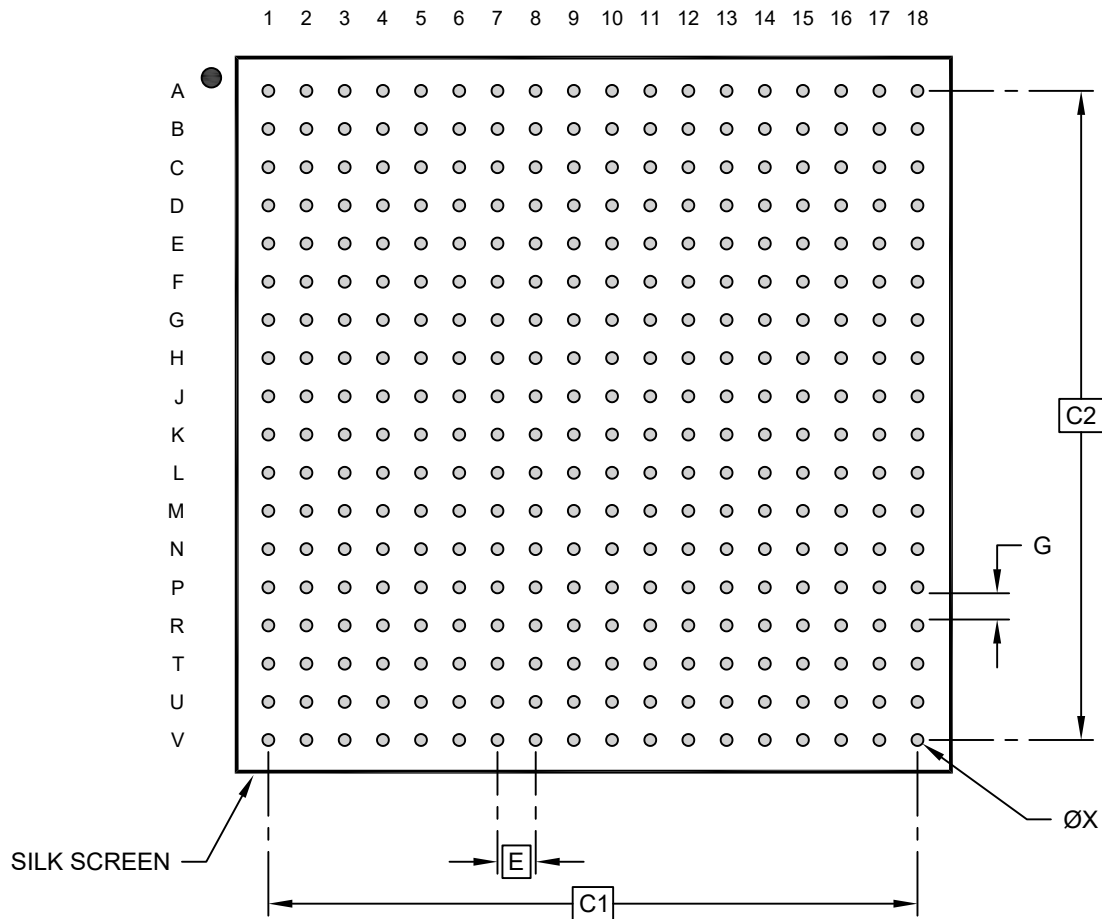
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	324		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.20
Ball Height	A1	0.16	0.21	0.26
Molded Package Thickness	A2	0.72	0.76	0.80
Mold Cap Thickness	A3	0.50 REF		
Overall Length	D	15.00 BSC		
Ball Array Length	D2	13.60 BSC		
Overall Width	E	15.00 BSC		
Ball Array Width	E2	13.60 BSC		
Ball Diameter	b	0.25	0.30	0.35

Notes:

1. Terminal A1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

324-Ball Thin Fine-Pitch Ball Grid Array (BJB) - 15x15x1.2 mm Body [TFBGA] Atmel Legacy Package CRC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	13.60 BSC		
Contact Pad Spacing	C2	13.60 BSC		
Contact Pad Width (Xnn)	X			0.25
Contact Pad to Contact Pad (Xnn)	G	0.55		

Notes:

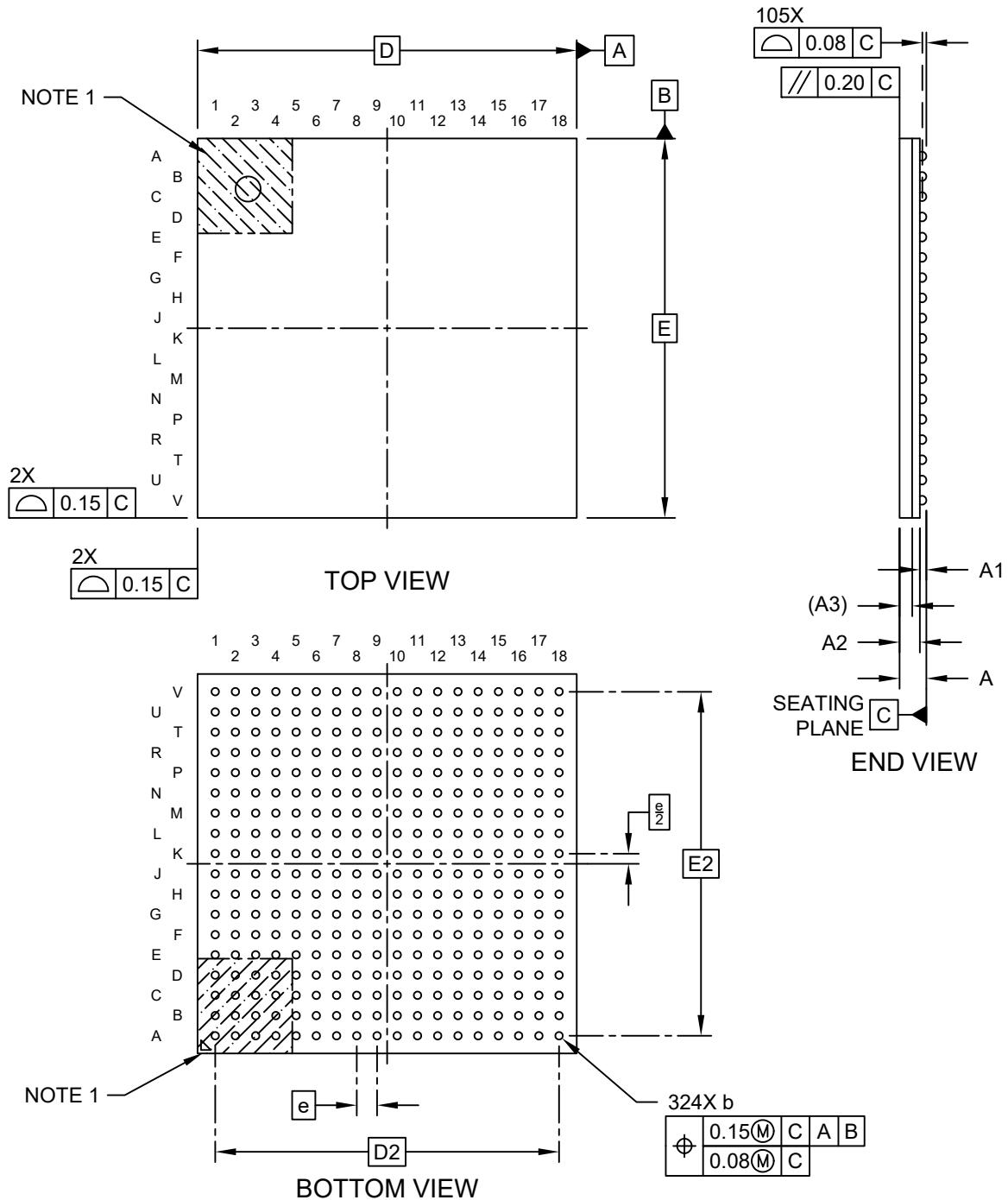
1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23148-BJB Rev A

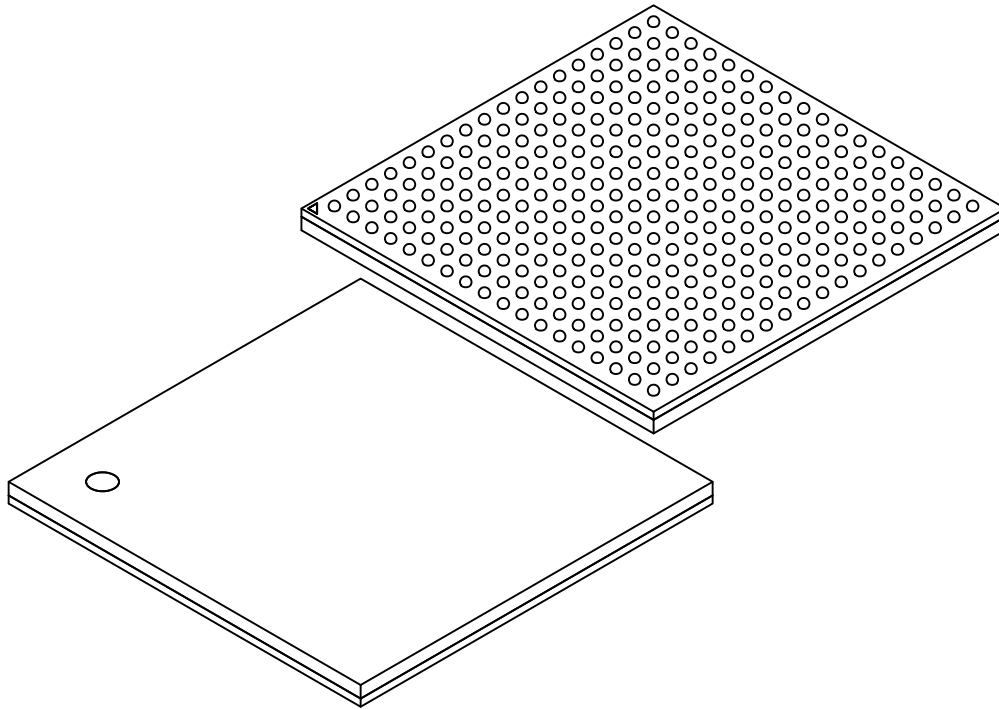
324-Ball Thin Fine-Pitch Ball Grid Array (BKB) - 15x15x1.2 mm Body [TFBGA] Atmel Legacy Package CRC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/package3>



324-Ball Thin Fine-Pitch Ball Grid Array (BKB) - 15x15x1.2 mm Body [TFBGA] Atmel Legacy Package CRC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



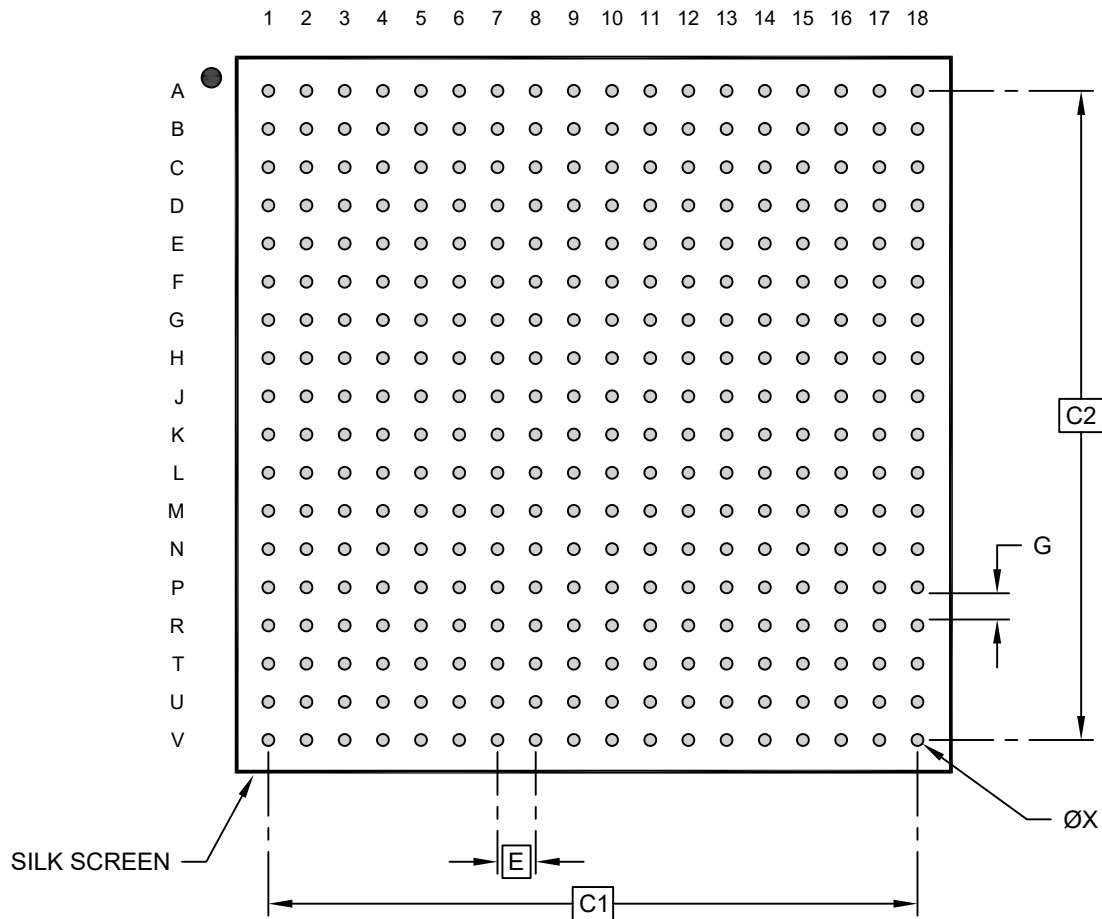
		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		324		
Pitch	e		0.80 BSC		
Overall Height	A		–	–	1.20
Ball Height	A1		0.16	0.21	0.26
Molded Package Thickness	A2		0.72	0.76	0.80
Mold Cap Thickness	A3		0.50 REF		
Overall Length	D		15.00 BSC		
Ball Array Length	D2		13.60 BSC		
Overall Width	E		15.00 BSC		
Ball Array Width	E2		13.60 BSC		
Ball Diameter	b		0.25	0.30	0.35

Notes:

1. Terminal A1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

324-Ball Thin Fine-Pitch Ball Grid Array (BKB) - 15x15x1.2 mm Body [TFBGA] Atmel Legacy Package CRC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	13.60 BSC		
Contact Pad Spacing	C2	13.60 BSC		
Contact Pad Width (Xnn)	X			0.25
Contact Pad to Contact Pad (Xnn)	G	0.55		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23148-BKB Rev A